

OBJECTIVE or PROBLEM

The goal of this project is to explore the use of augmented reality (AR) and computer vision technologies to reduce cognitive load during terminal block testing and to automatically generate digital artifacts that document wiring verification during testing.

OVERVIEW

- SEL manufactures protective relays and control equipment for power grid infrastructure. **They operate in a space where wiring accuracy is safety-critical**
- Current process: technician probes terminals with test probes while manually checking and annotating a wiring diagram.
 - Having to conduct two tasks at once with SEL's high accuracy and precision standards can cause mental and emotional fatigue
- Computer vision can detect probe positions overhead and a data acquisition unit (DAQ) can detect electrical continuity
 - Combining both enables automatic connection logging for verification and quality assurance
- Key goal from SEL: "demonstrate the approach's viability" as a working prototype**



Protective panels made by SEL

CONSTRAINTS

- The system shall run on a standard PC (Windows or Linux). No specialized embedded hardware or FPGA is required.
- The system shall not make any network calls to any cloud computing resources(must operate completely locally)
- Open source computer vision software (e.g. OpenCV) shall be used
- Results of circuits must be at least 95% accurate
- Project budget: \$500

APPROACH

Imaging: ArUco markers at panel corners lock perspective correction (homography) at startup

Detection: YOLO OBB (oriented bounding box) model identifies circuit components

Probe tracking: HSV color-range masking locates each probe tip in real time without the need of attached markers

Continuity: LabJack U12 polls for electrical continuity. Confirmed connection triggers logging to SQLite and AR overlay update

Record: At session end, a time stamped image with all AR overlays is saved as a digital verification record

HARDWARE DESIGN

The "eyes": Elgato Facecam 4K webcam

The "brain": Student's laptop

The "trigger": LabJack U12

The "arm": Microphone arm inserted into flange, which is attached directly into testing surface

The circuit: Connection devices given to the team by industry sponsor

SOFTWARE DESIGN

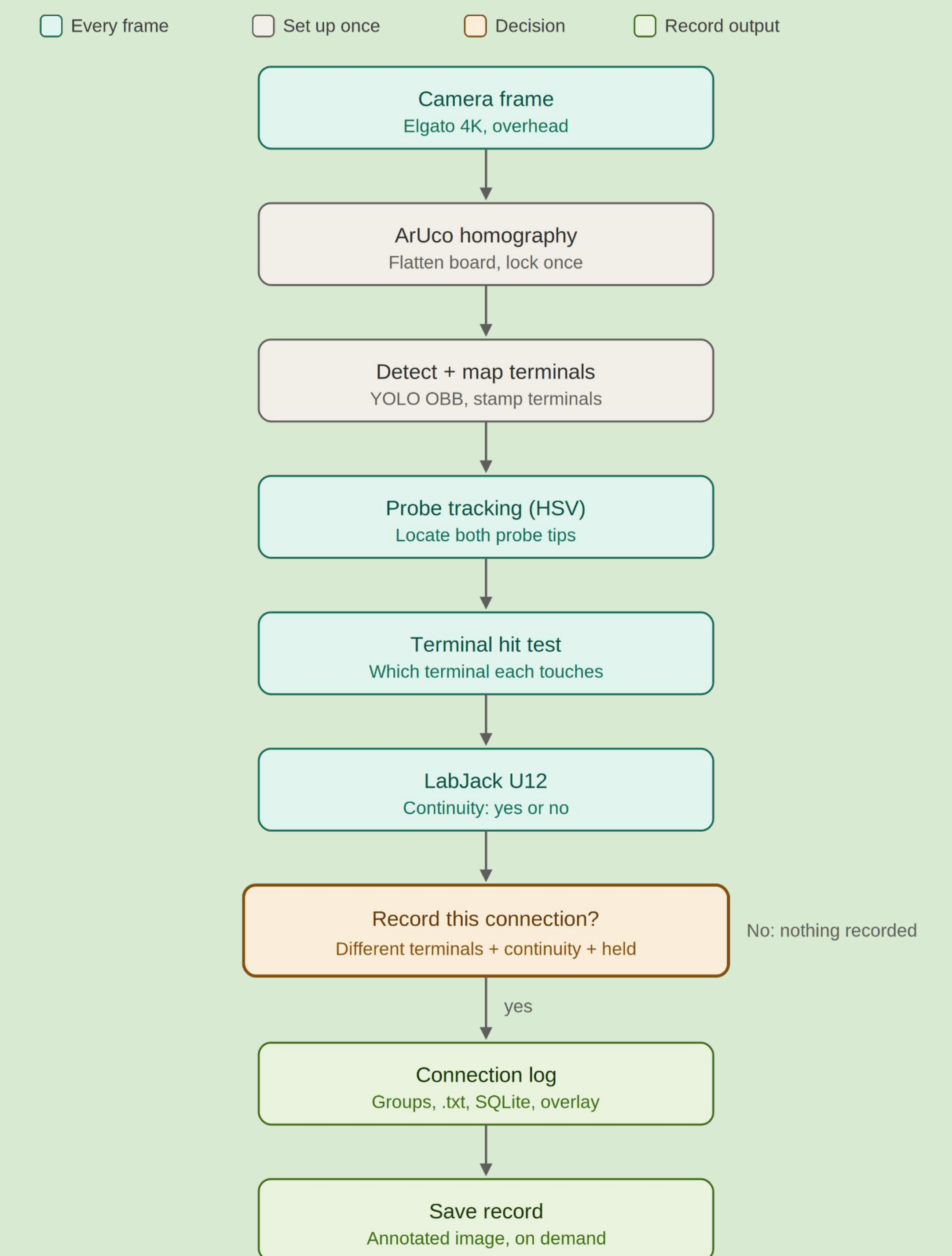
Decision-gated recording: a connection is logged only when the camera and the LabJack agree on two different, known terminals and the reading holds steady

Runs once vs. every frame: because the panel is fixed, detection and perspective correction lock once at startup. Only probe tracking and continuity run every frame, keeping the live view responsive.

Single source of truth: the list of recorded connections drives everything downstream: the on-screen groups, the text log, and the database. This ensures that they can never disagree, and "undo" just removes the last entry and rebuilds everything.

Markerless probe tracking: probe tips are located by color (HSV) with a quick click-to-teach step, so nothing has to be attached to the probes.

Honest about uncertainty: if the sensor drops out or a probe sits between terminals, the system shows a warning and records nothing rather than implying a result it can't stand behind.



Software Design Pipeline

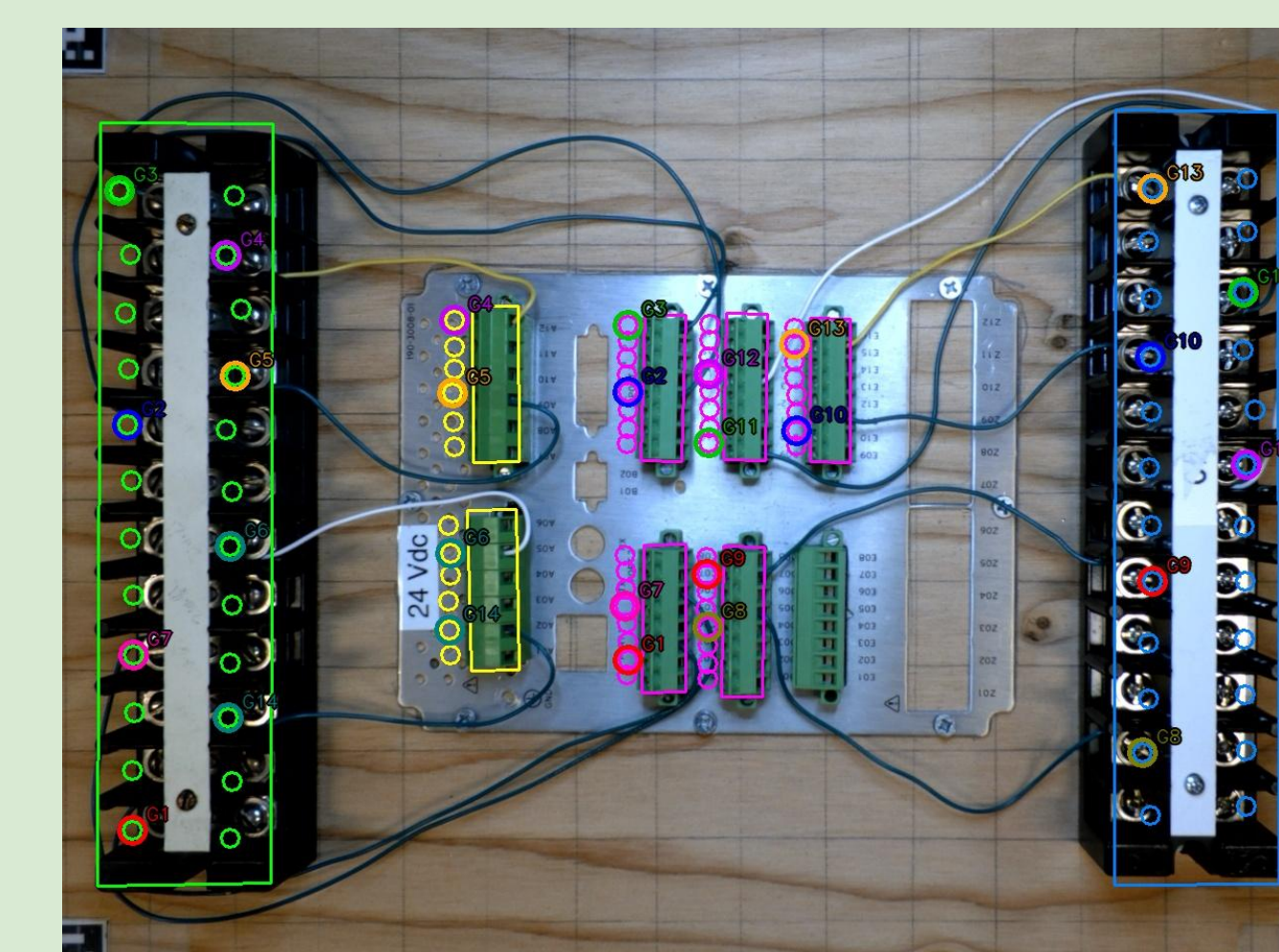
CONCLUSION, FINAL RESULTS and, FUTURE WORK

CONCLUSION

Viability demonstrated: the prototype successfully locates probe tips without physical markers and automatically logs verified continuity connections to time stamped text and database records

Reduces context switching by design: a technician probes terminals while the system automatically builds the annotation record, eliminating the need to switch attention between the circuit and a wiring diagram.

FINAL RESULT



Resulting annotated "circuit" board showing 13 of 14 test connections correctly

FUTURE WORK

Rotation-robust tip tracking: a Kalman filter to resolve the systematic tip-flip when a probe angles toward a frame edge.

Natural language queries: a local LLM generating read-only SQL queries against the inspection database for post-session analysis and quality assurance

AR headset interface: HoloLens 2 rendering spatial overlays anchored to the board via ArUco markers, with connection state broadcast from the Python backend over a local WebSocket connection.

Model fine-tuning: additional training images captured at the locked overhead position to improve confidence on partially occluded terminals.